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MILITARY SPECIFICATION

POWER CONTROL B: 10526480

1. SCOPE

1.1 This specification covers one type of power control circuit board, used in the Gun Direction Computer:M18. The power control B consists of warning detectors, kickout detectors and relay driver circuits.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invi tation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component parts

DRAWINGS

U.S. Army Munitions Command

F10526480	Power Control B

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

FSC 1220

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(Copies of specifications, standards, drawings, and packaging date sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The power control shall be manufactured in accordance with Drawing F10526480 and drawings pertaining thereto.

3.2 General requirements.- The power control shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The power control shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the power control for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the power control shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The power control shall withstand 3 shock impulses in each of 3 mutually perpendicular directions. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The power control shall meet the requirements of 3.4.2 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The power control shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80 F and +160 F.

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3.3.4 Humidity.

3.3.4.1 Operating.- The power control shall meet the requirements of 3.4.2 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135 F.

3.3.4.2 Non-operating.- The power control shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The power control shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The power control shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the power supply subassembly shall meet the performance requirements of this specification at standard ambient temperatures between 60 and 90°F and at a barometric pressure of 28 to 32 inches of mecury with a RH up to 90%.

3.4.1 Voltage detector diodes.

3.4.1.1 Forward voltage.- The forward voltage drop across each voltage detector diode (CR14, CR15, CR39, and CR40) shall be equal to or less than 0.6 volt direct current (vdc) at 1.5 ± 0.2 milliamperes (ma), forward current (If).

3.4.1.2 Leakage current.- The leakage current through each voltage detector diode shall be less than 20 microamperes (ua) at 15 ± 2 vdc back-bias.

3.4.1.3 Voltage detector static performance.- The components of voltage detector circuits shall possess the characteristics of 3.4.1.3.1 with the application of the following conditions:

(a) The following voltages shall be applied to the specified pins:

(1) -12 Vdc to pin 45A
(2) +35 Vdc to pin 32A
(3) +15 Vdc to pin 29A
(4) + 6 Vdc to pin 35A
(5) Variable supply (-17 Vdc to -33 Vdc) to pin 22A

(b) The dc returns of aforementioned voltage levels shall be connected to 38A.

(c) 470 ohms ±5%, 5W resistor connected from the collectors of the voltage limiters (Q10 and Q19) to pin 22A.

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(d) Pins 60A, 62A and 63B shorted to pin 38A.

(e) All voltages shall be relative to ground (Pin 38A).

3.4.1.3.1 The transistors shall be turned-off with a base-to-emitter voltage equal to or more positive than + 0.5 Vdc.

3.4.1.3.2 The transistors shall have a collector-to-emitter voltage of -0.15 ± 0.15 V in the saturated conditions.

3.4.1.3.3 The NPN transistor, Q9 shall have a collector-to-emitter voltage equal to or less than + 1.0 V in the saturated conditions.

3.4.1.3.4 The diode at the output of the emitter-follower shall be back biased 9 V minimum when the delay circuit output is 11.8 ± 0.5 Vdc.

3.4.1.3.5 Zener diodes shall have a zener voltage of 6.3 ± 0.6 V.

3.4.1.4 Relay driver diodes.

3.4.1.4.1 Forward voltage.- The forward voltage drop across the siodes shall not exceed 0.6 V at 1.5 ± 0.2 ma (If).

3.4.1.4.2 Leakage current.- The leakage current of the diode shall be less than 20 ua at 15 ± 2 V back-bias.

3.4.1.4.3 Relay driver static performance.- With the conditions specified in 3.4.1.3(a) (e), except that pins 62A and 63B shall not be shorted the characteristics of the components in the relay drivers shall be as follows:

(a) The transistor shall be turned off with base-to-emitter voltage equal to or greater than zero.

(b) The transistors shall have a collector-to-emitter voltage of -0.15 ± 0.15 V in the saturated condition.

(c) The transistors shall have a collector current, Ico, (with Ie = 0) of less than 40 ua.

3.4.1.5 Voltage inverter diodes.

3.4.1.5.1 Forward voltage.- The forward voltage drop across the diodes shall not exceed 0.6 volt at 1.5 ± 0.2 ma, (If).

3.4.1.5.2 Leakage current.- The leakage current of the diodes shall be less than 20 ua at 15 + 2 V back-bias.

3.4.1.5.3 Voltage inverter static performance.- The transistor (Q3) shall be turned OFF with a base-to-emitter voltage equal to or more positive than +0.5 V, when the conditions of 3.4.1.3(d) is met.

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3.4.1.5.4 The transistor (Q3) in the saturated conditions shall have a collector-to-emitter voltage of -0.15 ± 0.15 Vdc when pins 62A and 63B are shorted to pin 38A, and the connection between 60A and 38A is removed.

3.4.2 Circuit outputs.- The requirements of all the circuit outputs 3.4.2.1 through 3.4.2.10 shall be met when the conditions of 3.4.1.3(a) through (e) are applied.

3.4.2.1 High voltage warning.- The high voltage warning output shall switch from -11.0 ± 1.3 volts, direct current (Vdc) to -0.15 ± 0.15 Vdc, with a rise time of less than 2 microseconds (usec); when resistor R50 is within the range of 470 to 8.2 thousand (K) ohms and the input voltage at pin 22A is equal to or more negative than -29.5 Vdc. (Rise time shall be defined as that time duration which the amplitude of the leading edge is increasing from 10 to 90% of the maximum amplitude).

3.4.2.2 High voltage kickout.- The high voltage kickout output voltage (pin 43A) shall switch from -11.0 ± 1.3 Vdc to -0.15 ± 0.15 Vdc, with a rise time of less than 2 usec, when resistor R4 is within the range of 470 to 8.2K ohms and the input voltage at 22A is equal to or more negative than -31.5 Vdc.

3.4.2.3 Low voltage warning.- The low voltage warning output shall switch from -11.0 ± 1.3 Vdc to -0.15 ± 0.15 Vdc, with a rise time of less than 2 usec, when resistor R57 is within the range of 470 to 8.2 K ohms and the input voltage at pin 22A is equal to or more positive than -20.8 Vdc.

3.4.2.4 Low voltage transient kickout and delay.

a. The voltage at the collector of the trigger switch (Q6) shall switch from -6.9 ± 0.5 Vdc to minus 4.0 ± 0.6 Vdc with a rise time of less than 2 usec, when resistor R11 is within the range 1K to 22K ohms or 3.3K to 47K ohms and the input voltage at pin 22A is equal to or more positive than -18.8 Vdc.

b. The voltage at the collector of switch transistor, Q26 shall switch from -11.0 ± 0.5 Vdc to -0.15 ± 0.15 Vdc, 7 ± 3 sec after the voltage at trigger switch collector (Q6) has switched from -6.9 ± 0.5 Vdc to -4.0 ± 0.6 Vdc or when the input voltage at pin 22A is equal to or more positive than -18.8 Vdc. Resistor R45 shall be within the range of 3.3K to 47K ohms. The switching voltage of Q26 shall have a rise time of less than 2 usec.

NOTE: Resistors R11 and R45 are balancing resistors in the LVTK' circuit. R11 is determined for functional operation of the trigger network (3.4.2.4a) in the LVTK' circuit. R45 is determined during the final functioning of the LVTK' circuit (3.4.2.4b). If the delay time of 7 ± 3 seconds cannot be met, when R45 is within the range of 3.3K to 47K ohms; then R11 must be readjusted to comply to the range of 3.3K to 47K ohms.

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3.4.2.5 Low voltage kickout.- The voltage at the collector of the switch transistor (Q13) shall switch from -11.0 ± 1.3 Vdc to -0.15 to ± 0.15 Vdc with a rise time of less than 100 usec, when resistor R3 is within the range of 1K to 22K ohms and when the input voltage at pin 22A is equal to or more positive than 16.5 Vdc.

3.4.2.6 Error reset.- The voltage at the collector of switch transistor (Q27) shall switch from -0.15 ± 0.15 Vdc to -11.0 ± 1.3 Vdc with a fall time of less than 10 usec, when a -10 Vdc supply is momentarily connected to pin 2B and its return to pin 38A. The voltage at the collector of the switch transistor (Q28) shall switch from a -0.15 ± 0.15 Vdc to -11.0 ± 1.3 Vdc with a fall time of less than 10 usec, when a -10 Vdc supply is momentarily connected to pin 55B and its return to pin 38A.

3.4.2.7 Main power relay driver.- The voltage at the collector of Q10 shall switch from -27 ± 0.2 Vdc to -0.2 Vdc with a rise time of less than 2 usec, when the electrical connection between pin 63B and pin 38A is removed. The voltage at the collector of Q10 shall switch from -0.2 ± 0.2 Vdc to -27 ± 0.2 Vdc, with a fall time of less than 10 usec, when the electrical connection is reestablished between pin 63B and 38A. (Fall time shall be defined as that time duration during which the amplitude of the trailing edge is decreasing from 90 to 10% of the maximum amplitude).

3.4.2.8 Memory motor relay driver.- The voltage at the collector of Q19 shall switch from -27 ± 0.2 Vdc to -0.2 ± 0.2 Vdc with a rise time of less than 2 usec, when the electrical connection between pin 62A and pin 38A is removed. The voltage at the collector of Q19 shall switch from -0.2 ± 0.2 Vdc to -27 ± 0.2 Vdc with a fall time of less than 10 usec, when the electrical connection is reestablished between pin 62A and pin 38A.

3.4.2.9 Voltage warning inverter.- The voltage at the collector of the switch transistor (Q3) shall switch from -11.0 ± 1.3 Vdc to -0.15 ± 0.15 Vdc with a rise time of less than 2 usec, when the electrical connection between pin 60A and pin 38A is removed. The voltage at the collector of Q3 shall switch from -0.15 ± 0.15 Vdc to -11.0 ± 1.3 Vdc with a fall time of less than 10 usec when the electrical connection is reestablished between pin 60A and pin 38A.

3.4.3 Circuit outputs extreme temperatures.- At the temperatures specified in 3.3.3.1 and 3.3.4.1 the applicable circuit outputs (3.4.2.1 through 3.4.2.5) of the power control shall meet their respective requirements, except that the input voltage (pin 22A) may have a tolerance of ± 0.5 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein. components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II, and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60 to 90 F.)

TABLE I

(100 percent inspection)

Characteristic	Requirement	Test Procedures
101. Voltage detector diodes (vdd) forward voltage	3.4.1.1	4.5.2.2.1
102. Leakage current (vdd)	3.4.1.2	4.5.2.2.2
103. Voltage detector static performance	3.4.1.3 through 3.4.1.3.5	4.5.2.3 through 4.5.2.3.4
104. Relay driver diodes (rdd)- forward voltage	3.4.1.4.1	4.5.2.4.1
105. Leakage current (rdd)	3.4.1.4.2	4.5.2.4.2
106. Relay driver static performance	3.4.1.4.3(a) through (c)	4.5.2.4.3
107. Voltage inverter diodes (rid) forward voltage	3.4.1.5.1	4.5.2.5.1
108. Leakage current (rid)	3.4.1.5.2	4.5.2.5.2
109. Voltage inverter static performance	3.4.1.5.3, 3.4.1.5.4	4.5.2.5.3
110. High voltage warning	3.4.2.1	4.5.2.6.1
111. High voltage kickout	3.4.2.2	4.5.2.6.2

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TABLE I (Continued)

	Characteristic	Requirement	Test Procedures
112.	Low voltage warning	3.4.2.3	4.5.2.6.3
113.	Low voltage transient kickout and delay	3.4.2.4	4.5.2.6.4
114.	Low voltage kickout	3.4.2.5	4.5.2.6.5
115.	Error reset	3.4.2.6	4.5.2.6.6
116.	Main power relay driver	3.4.2.7	4.5.2.6.7
117.	Memory motor relay driver	3.4.2.8	4.5.9.9
118.	Voltage warning inverter	3.4.2.9	4.5.9.10

4.3.1 General. - One power control shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - Visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - Visual

4.3.2 Environmental. - Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity-Operating	3.3.4.1,3.4.2, 3.4.3	4.5.1.3.2,4.5.2,6 4.5.2.6.10
306.	Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4
308.	Operating temperature	3.3.3.1,3.4.2, 3.4.3	4.5.1.5,4.5.2,6 4.5.2.6.10
309.	Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure. - Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (See 4.2.3).

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4.4 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design. - The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment. - In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item Number	Description
1. Power Supply	Minus 12 vdc ± 2 percent, current rating greater than 80 ma., ripple less than 1.0 percent peak-to-peak, output impedance less than 2.0 ohms from dc to 500kc.
2. Power Supply	Plus 6 vdc ± 2 percent, current rating greater than 25 ma., ripple less than 1.0 percent peak-to-peak, output impedance less than 2.0 ohms from dc to 500kc.
3. Power Supply	Plus 15 vdc ± 2 percent, current rating greater than 12 ma., ripple less than 1.0 percent peak-to-peak, output impedance less than 2.0 ohms from dc to 500kc.
4. Power Supply	Plus 35 vdc ± 2 percent, current rating greater than 5 ma., ripple less than 1.0 percent peak-to-peak, output impedance less than 2.0 ohms from dc to 500kc.
5. Power Supply	Variable from minus 16 vdc to minus 33 vdc, resolution greater than 0.1 volt, current rating greater than 180 ma., ripple less than 0.1 percent peak-to-peak, output impedance less than 2.0 ohms from dc to 500kc.
6. Power Supply	Minus 10 vdc ± 2 percent, current rating greater than 5 ma., ripple less than 1.0 percent peak-to-peak, output impedance less than 2.0 ohms from dc to 500kc.

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7.	Voltage Measuring Device	D.C. accuracy ± 5 percent, voltage range 10mv to greater than or equal to 100K ohm.
8.	Oscilloscope	Accuracy ± 4 percent, rise time 50 nanoseconds or less, sweep shall be from 0.1 microseconds (us) per centimeter to 1 ms per centimeter or greater, input resistance greater than or equal to 1 megohm, input capacitance 20 pico farads less.
9.	Stop Watch	Dial registers in 1/2 seconds, one revolution of long hand every 60 seconds; accuracy ± 10%.
10.	Resistors(2)	470 ohm ± 5%, 5 watt.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration. - The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the power control rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock. - This test shall be conducted in accordance with the conditions of 3.3.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5G's. The power control shall be subsequently inspected for physical damage or defects in compliance with 3.3.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with the requirements of 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating. - This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating. - This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.6 to determine compliance with 3.3.4.1.

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4.5.1.4 Salt spray. - The power control shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2.6 for compliance with 3.3.6.

4.5.1.5 Temperature (Operating and storage). - After being subjected to all environmental tests specified above, the power control shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.2.6 to determine compliance with the requirements of 3.4.3 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2.1 to 4.5.9.10 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus. - To determine compliance with 3.3.5, one of the first 5 units produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance. - The following tests, except where otherwise specified, shall be conducted in conformance with 3.4.

4.5.2.1 Selection of resistors. - Prior to conducting the remaining tests herein, the selection of resistors shall be in accordance with the following procedures to insure that subsequent tests provide the purpose of determining compliance with 3.4. Resistors shall be selected for the high voltage warning, high voltage kickout, low voltage warning, low voltage transient kickout, and the low voltage kickout detectors such that the circuits shall operate at the applicable input voltages. Resistors shall be selected for the low voltage transient kickout delay such that a 7 ± 3 second delay time shall be achieved.

4.5.2.1.1 Input gates and power supplies. - Connect 63B and 62A to pin 38A and connect the pins of Power Control B with the respective power supplies as indicated in Table IV. The variable voltage supply (item 5 of 4.4.1.1) shall be initially set at minus 27.0 ± 0.5 vdc prior to connecting the power supplies to Power Control B.

TABLE IV

DC Power Supplies (volts)	Connect Power Supplies From Pin	To Pin
Minus 12, Item 1	45A	38A
Variable, Item 5	22A	38A
Plus 35, Item 4	32A	38A
Plus 15, Item 3	29A	38A
Plus 6, Item 2	35A	38A

4.5.2.1.2 High voltage warning. - Select a resistor from the list in Table V and install in the space provided for R50 in the Power Control B. The voltage of the variable supply item 5 shall be increased from the initial setting of minus 27.0 ± 0.5 vdc to minus 31 vdc. The voltage at pin 54B shall switch from minus 11.8 ± 0.5 vdc to minus 0.15 ± .15 vdc with a rise time of

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less than 2 microseconds while the voltage of the variable supply (Item 5 of 4.4.1.1) is increasing and is as close as possible to minus 29.5 vdc. Other resistance values listed in Table V shall be installed and the process repeated to determine the proper value for R50. The power supplies may have to be turned off and on to reset the voltage at pin 54B to minus 11.8 ± 0.5 vdc.

TABLE V

Resistance Values (Ohms)	Ordnance No.
470	10526085
560	10526086
680	10526087
698	10526088
750	10526089
820	10526740
1000	10526090
1240	10526091
1500	10526092
1800	10526093
2200	10526094
2700	10526095
3300	10525856
3570	10526698
3900	10526096
4320	10526097
4700	10526098
5600	10526099
6800	10526100
8200	10526028

4.5.2.1.3 High voltage kickout.- Select a resistor from Table V and install in the space provided for R4 in the Power Control B. The Voltage of the variable supply item 5, shall be increased from the initial setting of minus 27.0 ± 0.5 vdc to minus 33 vdc. The voltage at pin 43A shall switch from minus 11.8 ± 0.5 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 2 microseconds while the voltage of the variable supply item 5, is increasing and is as close as possible to minus 31.5 vdc. Other resistance values listed in Table V shall be installed and the process repeated to determine the proper value for R4. The power supplies may have to be turned off and on, to reset the voltage at pin 43A to minus 11.8 ± 0.5 vdc.

4.5.2.1.4 Low voltage warning.- Select a resistor from the list in Table V and install in the space provided for R57 in the Power Control B. The voltage of the variable supply (item 5 of 4.4.1.1) shall be decreased from the initial setting of minus 27.0 ± 0.5 vdc to minus 20 vdc. The voltage at pin 1B shall switch from minus 11.8 ± 0.5 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 2 microseconds while the voltage of the variable supply item 5, is decreasing and is as close as possible to minus 20.8 vdc. Other resistance values from Table V shall be installed and the process repeated to determine the proper value for R57. The power supplies may have to be turned off and on to reset the voltage at pin 1B to minus 11.8 ± 0.5 vdc.

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4.5.2.1.5 Low voltage transient kickout.- Select a resistor from the list in Table VI and install in the space provided for R11 in the Power Control B. The voltage of the variable supply (item 5 of 4.4.1.1) shall be decreased from the initial setting of minus 27.0 ± 0.5 vdc to minus 18.5 vdc. The voltage at pin 26A shall switch from minus 6.9 ± 0.5 vdc to minus 4.0 ± 0.6 vdc with a rise time of less than 2 microseconds while the voltage of the variable supply is decreasing and is as close as possible to minus 18.8 vdc. Other resistance values from Table VI shall be installed and the process repeated to determine the proper value for R11.

TABLE VI

Resistance Values (Ohms)	Ordnance No.
1000	10526090
1240	10526091
1500	10526092
1800	10526093
2200	10526094
2700	10526095
3300	10525856
3900	10526096
4700	10526098
5600	10526099
6800	10526100
8200	10526028
10000	10526599
12000	10526032
15000	10526033
18000	10526042
22000	10526044
3570	10526698
4320	10526097
9100	10526598
16900	10526697
20000	10526043

4.5.2.1.6 Low voltage kickout.- Select a resistor from the list in Table VI and install in the space provided for R3 in the Power Control B. The voltage of the variable supply item 5, shall be decreased from the initial setting of minus 27.0 ± 0.5 vdc to minus 16 vdc. The voltage at pin 23B shall switch from minus 11.8 ± 0.5 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 100 microseconds while the voltage of the variable supply item 5, is decreasing and is as close as possible to minus 16.5 vdc. Other resistance values from Table VI shall be installed and the process repeated to determine the proper value for R3. The power supplies may have to be turned off and on to reset the voltage at pin 23B to minus 11.8 ± 0.5 vdc.

4.5.2.1.7 Low voltage transient kickout delay.- The selected resistor for R11 must be installed for this test. Select a resistor from the list in Table VII and install in the space provided for R45 in the Power Control B.

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The voltage of the variable supply item 5, shall be decreased from the initial setting of minus 27.0 ± 0.5 vdc until the voltage at pin 26A switches from minus 6.9 ± 0.5 vdc to minus 4.0 ± 0.6 vdc with a rise time of less than 2 microseconds. The voltage at pin 40B shall switch from minus 11.8 ± 0.5 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 2 microseconds and delay time of 7 ± 3 seconds. The delay time shall be measured from the time the voltage at pin 26A switched from minus 6.9 ± 0.5 to minus 4.0 ± 0.6 vdc. Other resistance values from Table VII shall be installed in the space provided for R11 and the process repeated if the delay time is not as specified. All test leads and power supplies shall be disconnected upon completion of this test.

TABLE VII

Resistance Values (Ohms)	Ordnance No.
3300	10525856
3900	10526096
4700	10526098
5600	10526099
6800	10526100
8200	10526028
10000	10526599
12000	10526032
15000	10526033
18000	10526042
22000	10526044
27000	10526898
33000	10526638
39000	10526899
47000	10526029
3570	10526698
4320	10526097
9100	10526598
16900	10526697
20000	10526043

4.5.2.2 Voltage detector diodes.- The following test shall be conducted to determine compliance with the requirements of 3.4.1.

4.5.2.2.1 Forward voltage drop.- The minus 12 vdc power supply (item 1 of 4.4.1.1) shall be connected to the pins of Power Control B listed in column A of Table VIII. The minus 12 vdc power supply return shall be connected through a 8.2K ± 5 percent ½ watt resistor to the respective pins of Power Control B listed in Column B of Table V. The voltages shall be as listed in column C of Table VIII when measured from pins of column B to its repsective pins in column A todetermine compliance with 3.4.1.1.

TABLE VIII

Column A Pin	Column B Pin	Column C (vdc)
2B	TP14	Plus 4.8 ± 1.0
55B	51A	Plus 4.8 ± 1.0
30B	29B	Plus 1.8 maximum

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4.5.2.2.2 Leakage current.- The plus 15 vdc power supply item 3, shall be connected to pins of Power Control B listed in column A of Table IX. The plus 15 vdc power supply return shall be connected through a 12K ±5 percent ½ watt resistor to the pins of Power Control B listed in column B of Table IX. To determine compliance with 3.4.1.2, the voltages shall be less than the values listed in column C of Table IX when measured from the pins of column B of Table IX to the plus 15 vdc power supply return. All test leads shall be disconnected upon completion of this test.

TABLE IX

Column A Pin	Column B Pin	Column C (m vdc)
2B	TP 14	Plus 240
55B	51A	Plus 240

4.5.2.3 Voltage detector static test.- To determine complicance with the voltage applications and conditions specified in 3.4.1.3(a) through (e) the following tests shall be conducted and the measurements shall be performed in compliance with the requirements of 3.4.1.3.1 through 3.4.1.3.5.

4.5.2.3.1 Resistors of 470 ohms, ± 5 percent, 5 watt shall be connected from pin 51B to pin 22A and from pin 58B to pin 22A. A connection shall be made from pins 60A, 62A and 63B to pin 38A. the dc power supplies listed in Table IV shall be connected to the indicated pins. The variable voltage supply (item 5 of 4.4.1.1) shall be initially set at minus 27 ± 0.5 vdc prior to connecting the power supplies to the Power Control B. The voltages measured at pins 23B, 1B, 40B, 43A, and 54B with respect to pin 38A shall be minus 11.8 ± 0.5 vdc. If the voltages at pins 23B, 1B, 40B, 43A, and 54B are not as specified turn the power supplies off and on again to reset the voltage at the specified pins, then make the measurements indicated in Table X.

TABLE X

Measure from Pin	To Pin	Measurement (vdc)
TP10	38A	plus 0.5 minimum
4A	38A	plus 0.5 minimum
2A	38A	plus 0.5 minimum
25A	24A	plus 0.5 minimum
39B	38A	plus 0.5 minimum
49A	38A	plus 0.5 minimum
TP3	38A	plus 0.5 minimum
52A	38A	plus 0.5 minimum
TP6	38A	plus 0.5 minimum
TP16	38A	plus 0.5 minimum
22A	22B	minus 12.4 ± 0.8
10A	38A	minus 0.15 ± 0.15
11A	38A	minus 0.15 ± 0.15
6B	38A	minus 0.15 ± 0.15
3A	38A	minus 0.15 ± 0.15
TP2	24A	minus 0.15 ± 0.15
39B	35A	minus 0.15 ± 0.15

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TABLE X (COND)

Measure from Pin	To Pin	Measurement (vdc)
TP7	38A	minus 0.15 ± 0.15
54A	38A	minus 0.15 ± 0.15
28B	38A	minus 6.2 ± 0.4
19A	38A	minus 7 ± 1
3B	38A	minus 7 ± 1
43B	38A	minus 7 ± 1
52B	38A	minus 7 ± 1
30B	26A	plus 1.0 maximum

4.5.2.3.2 The voltage of the variable supply (item 5 of 4.4.1.1) shall be varied from the initial setting of minus 27.0 ± 0.5 vdc toward minus 33 vdc until the voltage at pin 43A switches from -11.8 ± 0.5 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 2 microseconds. The measurements of Table XI shall be performed.

TABLE XI

Measure from Pin	To Pin	Measurement (vdc)
TP4	38A	plus 0.5 vdc minimum
51A	38A	plus 0.5 minimum
43B	38A	minus 0.15 ± 0.15
52B	38A	minus 0.15 ± 0.15
22A	49B	minus 24.0 ± 1.2
TP7	38A	minus 7 ± 1
54A	38A	minus 7 ± 1

4.5.2.3.3 The voltage of the variable supply (item 5 of 4.4.1.1) shall be varied from the initial setting toward minus 16 vdc until the voltage at pin 23B switches from minus 11,8 ± 0.5 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 2 microseconds. After allowing 10 seconds or more for the circuits to settle in its stable conditions the measurements in Table XII shall be performed.

TABLE XII

Measure from Pin	To Pin	Measurements (vdc)
TP5	38A	plus 0.5 minimum
TP14	38A	plus 0.5 minimum
35B	35A	plus 0.5 minimum
19A	38A	minus 0.15 ± 0.15
3B	38A	minus 0.15 ± 0.15
26A	24A	minus 0.15 ± 0.15
29A	30A	plus 1.0 maximum
43A	43B	plus 0.8 maximum
54B	52B	plus 0.8 maximum
26A	28B	plus 0.5 minimum

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4.5.2.3.4 The voltage of the variable supply (item 5 of 4.4.1.1) shall be reset to minus 27.0 ± 0.5 vdc. The measurements in Table XIII shall be performed. All test leads and power supplies shall be disconnected upon completion of these measurements.

TABLE XIII

Measure from Pin	To Pin	Measurements (vdc)
23B	19A	plus 0.8 maximum
1B	3B	plus 0.8 maximum

4.5.2.4 Relay driver diodes.- Compliance with 3.4.1.4 shall be determined by conducting the following tests.

4.5.2.4.1 Forward voltage drop.- The minus 12 vdc power supply (item 1 of 4.4.1.1) shall be connected to pins of Power Control B listed in column A of Table XIV. The minus 12 vdc power supply return shall be connected through a 8.2K ohms ±5 percent ½ watt resistor to the respective pins of Power Control B listed in column B of Table XIV. The voltages shall be as listed in column C of Table XIV when measured from pins of column B to the respective pins of column A to determine compliance with 3.4.1.4.1.

TABLE XIV

Column A Pin	Column B Pin	Column C (vdc)
63A	65A	plus 0.6 maximum
63A	65B	plus 0.6 maximum
63A	64A	plus 0.6 maximum
63A	64B	plus 0.6 maximum
63A	63B	plus 0.6 maximum
51B	22A	plus 0.8 maximum
58B	22A	plus 0.8 maximum
38A	57B	plus 0.6 maximum
38A	61B	plus 0.6 maximum
59B	61B	plus 2.2 ± 1.0
55A	57B	plus 2.2 ± 1.0

4.5.2.4.2 Leakage current.- The plus 15 vdc power supply (item 3 of 4.4.1.1) shall be connected to the pins of Power Control B listed in column A of Table XV. The plus 15 vdc power supply return shall be connected through a 12K ohms ± 5 percent ½ watt resistor to the pins of Power Control B of Table XV listed in column B of Table XV to determine compliance with 3.1.4.2 the voltages shall be less than the values listed in column C of Table XV when measured from the pins of column B of Table XV to the plus 15 vdc power supply return. All test leads shall be disconnected upon completion of measurements.

TABLE XV

Column A Pin	Column B Pin	Column C (mvdc)
63A	65A	plus 240
63A	65B	plus 240
63A	64A	plus 240
63A	64B	plus 240
63A	63B	plus 240

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4.5.2.4.3 Relay driver static test.- The following tests shall be conducted to determine compliance with 3.4.1.4.3 and the conditions specified therein. An electrical connection shall be made from pins 63B and 62A to pin 38A. A resistor of 470 ohms ± 5 percent 5 watt shall be connected from pin 51B to pin 22A and from pin 58B to pin 22A. The dc power supplies listed in Table IV shall be connected to the indicated pin of Power Control B. The variable voltage supply (item 5 of 4.4.1.1) shall be initially set to minus 27 ± 0.1 vdc prior to connecting the power supplies to the Power Control B. The measurements of Table XVI shall be performed. Remove the connections between pin 63B to 38A, and 62A to pin 38A. The measurements of Table XVII shall be performed.

TABLE XVI

Measure from	To Pin	Measurements
TP11	38A	minus 13.5 ± 2 vdc
TP15	38A	minus 13.5 ± 2 vdc
TP8	38A	minus 13.5 ± 2 vdc
TP13	38A	minus 13.5 ± 2 vdc
59B	63A	more negative than plus 0.2 vdc
57B	38A	plus 0.35 ± .25 vdc
61B	38A	plus 0.35 ± .25 vdc
59A	45A	less than plus 40 mv
TP12	45A	less than plus 40 mv

TABLE XVII

Measure from Pin	To Pin	Measurements (vdc)
TP8	38A	minus 0.15 ± 0.15
TP13	38A	minus 0.15 ± 0.15
58B	TP8	minus 0.15 ± 0.15
51B	TP13	minus 0.15 ± 0.15
TP12	59B	minus 0.15 ± 0.15
59A	55A	minus 0.15 ± 0.15

4.5.2.5 Voltage inverter diodes.- Compliance with 3.4.1.5 shall be determined by the following tests.

4.5.2.5.1 Forward voltage drops.- The minus 12 vdc power supply (item 1 of 4.4.1.1) shall be connected to the pins of Power Control B listed in column A of Table XVIII. The minus 12 vdc power supply return shall be connected through a 8.2K ohms ± 5 percent ½ watt resistor to the respective pins of Power Control B, listed in column B of Table XVIII. The voltage, shall not exceed the values listed in column C of Table XVIII when measured from pins of column B to the respective pins of column A to determine compliance with 3.4.1.5.1.

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TABLE XVIII

Column A Pin	Column B Pin	Column C (vdc)
58A	60A	plus 0.6
58A	60B	plus 0.6

4.5.2.5.2 Leakage current.- The plus 15 vdc power supply (Item 3 of 4.4.1.1) shall be connected to the pins of Power Control B listed in column A of Table XIX. The plus 15 vdc power supply return shall connected through a 12K ohms ± 5 percent ½ watt resistor to the pins of Power Control B listed in column B of Table XIX. The voltages shall be less than the values listed in column C of Table XIX when measured from the pins of column B of Table XIX to the plus 15 vdc power supply return to determine compliance with 3.4.1.5.2.

TABLE XIX

Column A Pin	Column B Pin	Column C (mvdc)
58A	60A	Plus 240
58A	60B	Plus 240

4.5.2.5.3 Voltage inverter static test.- To determine compliance with 3.4.1.5.3 and 3.4.1.5.4 connect pins 63B 60A and 62A to pin 38A and measure the direct current voltages (vdc) specified in Table XX. After testing accordance with Table XX, remove the connection between pins 60A and 38A and then measure the voltage at pin 62B with respect to pin 38A. The measurement shall be more positive than minus 0.3 vdc.

TABLE XX

Measure from Pin	To Pin	Measurements (vdc)
TP1	38A	plus 0.5 minimum
62B	38A	minus 11.8 ± 0.5

4.5.2.6 Circuit outputs.- The requirements and conditions specified in 3.4.2 shall be determined by the tests outlined below. Resistors of 470 ohms ± 5 percent, 5 watt shall be connected from pin 58B to pin 22A and from pin 51B to pin 22A. An electrical connection shall be made from pins 60A, 62A, 63B to pin 38A. The dc power supplies of Table I shall be connected to the indicated pins of Power Control B. The variable voltage supply (item 5 of 4.4.1.1) shall be initially set at minus 27 ± 0.5 vdc prior to connecting the power supply to the Power Control B. All voltage measurements shall be made with respect to pin 38A.

4.5.2.6.1 High voltage warning.- To determine compliance with 3.4.2.1, the voltage at pin 54B shall switch from minus 11.0 ± 1.3 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 2 microseconds when the voltage of the variable supply (item 5 of 4.4.1.1) is varied from minus 27 ± 0.5 vdc to minus 29.5 ± 0.2 vdc.

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4.5.2.6.2 High voltage kickout.-To determine compliance with 3.4.2.2 the volt at pin 43A shall switch from minus 11.0 ± 1.3 vdc to minus 0.15 vdc with a rise time of less than 2 microseconds when the voltage of the variable supply (item 5 of 4.4.1.1) is varied from minus 29.5 ± 0.2 vdc to minus 31.5 ± 0.2 vdc.

4.5.2.6.3 Low voltage warning.- To determine compliance with 3.4.2.3 voltage at pin 1B shall switch from minus 11.0 ± 1.3 vdc to minus 0.15 ± .15 vdc with a rise time of less than 2 microseconds when the voltage of the variable supply (Item 5 of 4.4.1.1) is varied from minus 31.5 ± 0.2 vdc to minus 20.8 ± 0.2 vdc.

4.5.2.6.4 Low voltage transient kickout and delay.- To determine compliance with 3.4.2.4(a) and (b) the voltage at pin 26A shall switch from minus 6.9 ± 0.5 vdc to minus 4.0 ± 0.6 vdc with a rise time of less than 2 microseconds when the voltage of the variable supply specified as varied from minus 20,8 ± 0.2 vdc to minus 18.8 ± 0.2 vdc. The voltage at pin 40B shall switch from minus 11.0 ± 1.3 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 2 microseconds and a delay time of 7 ± 3 seconds. The delay time shall be measured from the time the voltage at pin 26A switched from minus 6.9 ± 0.5 vdc to minus 4.0 ± 0.6 vdc.

4.5.2.6.5 Low voltage kickout.- To determine compliance with 3.4.2.5 the voltage at pin 23B shall switch from minus 11.0 ± 1.3 to minus 0.15 ± 0.15 vdc with a rise time of less than 100 microseconds when the voltage of the variable supply (item 5 of 4.4.1.1) is varied from minus 18.8 ± 0.2 vdc to minus 16.5 ± ± 0.2 vdc. The voltage of the variable supply shall then be varied from minus 18.8 to minus 27 ± 0.1 vdc. The voltage measured at pins 1B 23B, 43A and 54B shall be minus 0.15 ± 0.15 vdc. The voltage measured at pin 26A shall be minus 6.5 ± 0.5 vdc.

4.5.2.6.6 Error reset.- To determine compliance with 3.4.2.6 the voltage at pin 1B shall switch from minus 0.15 ± 0.15 vdc to minus 11.0 ± 1.3 vdc with a fall time of less than 10 microseconds when the minus 10 vdc power supply (item 6 of 4.4.1.1) is momentarily connected to pin 2B with the minus 10 vdc power supply return connected to pin 38A. The voltage at pin 54B shall switch from minus 0.15 ± 0.15 vdc to minus 11.0 ± 1.3 vdc with a fall time of less than 10 microseconds when the minus 10 vdc power supply (item 6 of 4.4.1.1) is momentarily connected to pin 55B with the minus 10 vdc power supply return connected to pin 38A. Fall time shall be defined as that time duration during which the amplitude of the trailing edge is decreasing from 90 to 10 percent of the maximum amplitude.

4.5.2.6.7 Main power relay driver.- To determine compliance with 3.4.2.7, the voltage at pin 58B shall switch from minus 27 ± 0.2 vdc to minus 0.2 ± 0.2 vdc with a rise time of less than 2 microseconds when the electrical connection between pin 63B to pin 38A is removed. The voltage at pin 58B shall switch from minus 0.2 ± 0.2 vdc to minus 27 ± 0.2 vdc with a fall time of less than 10 microseconds when the electrical connection is re-established between pin 63B to pin 38A.

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4.5.2.6.8 Memory motor relay driver.- To determine compliance with 3.4.2.8, the voltage at pin 51B shall switch from minus 27 ± 0.2 vdc to minus 0.2 ± 0.2 vdc with a rise time of less than 2 microseconds when the electrical connection between pin 62A to pin 38A is removed. The voltage at pin 51B shall switch from minus 0.2 ± 0.2 vdc to minus 27 ± 0.2 vdc with a fall time of less than 10 microseconds when the electrical connection is re-established between pin 62A to pin 38A.

4.5.2.6.9 Voltage warning inverter.- To determine compliance with 3.4.2.9, the voltage at pin 62B shall switch from minus 11.0 ± 1.3 vdc to minus 0.15 ± 0.15 vdc with a rise time of less than 2 microseconds when the electrical connection between pin 60A to pin 38A is removed. The voltage at pin 62B shall switch from minus 0.15 ± 0.15 vdc to minus 11.0 ± 1.3 vdc with a fall time of less than 10 microseconds when the electrical connection is re-established between pin 60A to pin 38A.

4.5.2.6.10 Circuit output, extreme temperatures.- Above tests specified in 4.5.2.6 through 4.5.2.6.9 shall be performed at standard ambient temperatures and at the extreme temperatures specified in 4.5.1.3.2 and 4.5.1.5, except the voltage tolerances at pin 22A specified in paragraphs 4.5.2.6.1 thru 4.5.2.6.5 shall be increased from ± 0.2 vdc to ± 0.5 vdc for the extreme temperature tests.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The power control B consists of; and is intended to do the following:

(a) High voltage kickout detector: Gives an output signal when the input signal exceeds the normal high or safe range.

(b) Low voltage kickout detector: Gives an output signal when the input voltage is below the normal low or safe range. Also gives an output signal when all the three phases of the input voltage are not present.

(c) Low voltage warning detector: Gives an output signal when the input signal is below the normal low operating range.

(d) High voltage warning detector: Gives an output signal when the input signal exceeds the normal high operating range.

(e) Low voltage transient kickout: Gives an output signal when the voltage source is overloaded and the input voltage is below a specified limit for a nominal seven seconds.

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(f) Voltage warning inverter: Gives a true output signal when one or both of the inverter inputs is false.

(g) Main power relay driver: Activates a relay to apply the power to the computer.

(h) Memory motor relay driver: Activates a relay to apply the power to the memory motor when the memory motor temperature is above a specific limit.

6.2 Ordering Data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (see 5.1).
(e) A block of serial numbers to identify each power control.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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